Problem 1a) (15 points)

Given the image region as shown in Figure 1 and $V = \{1\}$, what is the shortest m-path between p (the pixel at the upper-left corner) and q (the pixel at the bottom-right corner)? Give the length of the path if the path exists.

$$\begin{matrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 \end{matrix}$$

Step 1: Identify pixels in V

p

1	0	1	1	0
0	1	0	1	0
0	0	1	1	0
1	1	0	1	0
0	1	1	1	1

q

All pixels with value 1 are highlighted. These are the only candidates for the m-path since $V = \{1\}$.

Step 2: Apply m-adjacency rules

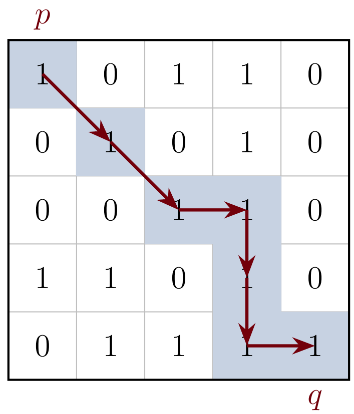
Two pixels with values in V are **m-adjacent** if:

a) They are 4-adjacent (share an edge), OR

b) They are diagonally adjacent AND their shared 4-neighbors contain no pixels with values in V.

The second condition prevents diagonal shortcuts when a 4-connected bridge exists, eliminating path ambiguity.

Step 3: Trace the shortest m-path



Path: $(0,0) \to (1,1) \to (2,2) \to (2,3) \to (3,3) \to (4,3) \to (4,4)$

Diagonal move justifications:

- $(0,0) \to (1,1)$: Shared 4-neighbors are $(0,1) = 0$ and $(1,0) = 0$. Neither is in V, so m-adjacent.
- $(1,1) \to (2,2)$: Shared 4-neighbors are $(1,2) = 0$ and $(2,1) = 0$. Neither is in V, so m-adjacent.

Results

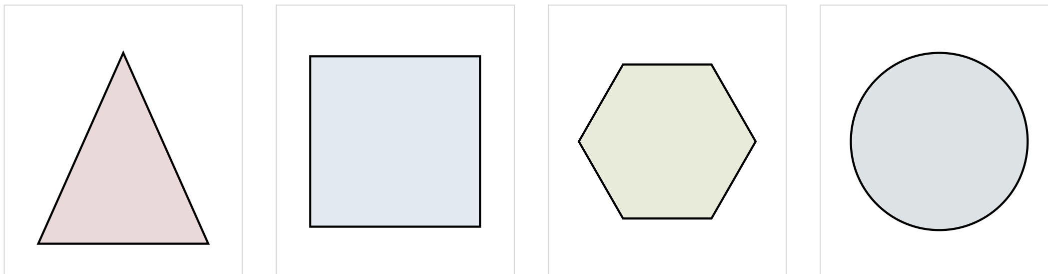
The shortest m-path exists and has **length 6**.

Problem 1b) (15 points)

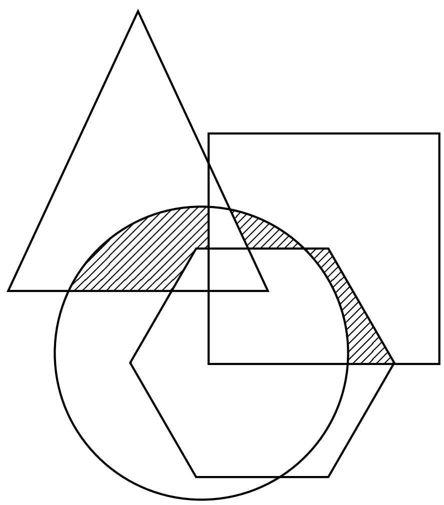
To take a picture for fireworks at night, give at least two operations that should be applied to improve the visualization quality of the image and explain.

Problem 2) (20 points)

A, B, C and D are four sets. Give expressions for the set shown shaded in Figure 2 in terms of A, B, C, and D. Note that all of the shaded areas constitute of one set



A B C D



Problem 3) (25 points)

Given a 3x3 spatial filter $\frac{1}{9}\begin{bmatrix}-1 & -1 & -1\\ -1 & -17 & -1\\ -1 & -1 & -1\end{bmatrix}$ answer the following questions:

a) (10 points) What type of filter is it? When applying this filter on an image, what kind of effect you will expect in the output image? Why? (Hint: is this filter an image smoothing filter, an image sharpening filter or an edge detector?)

b) (15 points) Perform the image convolution using this filter on a 3x3 image $\begin{bmatrix}3 & 5 & 12\\ 7 & 18 & 11\\ 5 & 9 & 15\end{bmatrix}$.

Give the CROPPED filtering result for the output image. (Hint: you need to assume an appropriate zero mapping)

Problem 4) (25 points)

An image with intensities in the range $r \in [0, L-1]$ has the pdf

$$p_r(r) = \begin{cases} \dfrac{4r}{3(L-1)^2} & 0 \leq r \leq L-1 \\ 0 & \text{otherwise} \end{cases}$$

where $L-1$ is the maximum intensity value. It is desired to transform the intensity levels of this image so that they will have the specified pdf

$$p_z(z) = \begin{cases} \dfrac{2z}{(L-1)^2} & 0 \leq z \leq L-1 \\ 0 & \text{otherwise} \end{cases}$$

Assume that the intensity value is continuous and find the transformation function $z = T(r)$ in terms of r and z that will accomplish this.

(Hints: The transformation function of histogram equalization is given as

$$s = T(r) = (L-1) \int_0^r p_r(w)\, dw$$

)

Bonus Question (20 points)

Perform the Fourier transform to a function as shown in Figure 3.

(Hint: the definition of the Fourier transform of a function is $F(u) = \int_{-\infty}^{\infty} f(t)e^{-j2\pi ut}\, dt$)